

## Quiz (Carolina Reaper)

**Q1.** What is the formula for solving  $ax^2 + bx + c = 0$ ?

- (A)  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$   
 (B)  $x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$   
 (C)  $x = \frac{-b \pm \sqrt{b^2 - 2ac}}{2a}$   
 (D)  $x = \frac{-b \pm \sqrt{b^2 + 2ac}}{2a}$   
 (E)  $x = \frac{-b \pm \sqrt{b^2 - ac}}{2a}$

(E) None of the above

**Q2.** What is the discriminant in the quadratic formula?

- (A)  $b^2 - 4ac$   
 (B)  $b^2 + 4ac$   
 (C)  $b^2 - 2ac$   
 (D)  $b^2 + 2ac$   
 (E)  $b^2 - ac$

**Q5.** If the discriminant is negative, how many solutions does the quadratic equation have?

- (A) 1  
 (B) 2  
 (C) 0  
 (D) Infinite  
 (E) None of the above

**Q3.** If the discriminant is positive, how many solutions does the quadratic equation have?

- (A) 1  
 (B) 2  
 (C) 0  
 (D) Infinite  
 (E) None of the above

**Q6.** Solve  $x^2 + 4x + 4 = 0$  using the quadratic formula.

- (A)  $x = -2$   
 (B)  $x = -2, x = 2$   
 (C)  $x = 0$   
 (D)  $x = 2$   
 (E)  $x = -4$

**Q4.** If the discriminant is zero, how many solutions does the quadratic equation have?

- (A) 1  
 (B) 2  
 (C) 0  
 (D) Infinite

**Q7.** Solve  $x^2 - 7x + 12 = 0$  using the quadratic formula.

- (A)  $x = 3, x = 4$   
 (B)  $x = -3, x = -4$   
 (C)  $x = 0$   
 (D)  $x = 7$   
 (E)  $x = 12$

**Q8.** What is the value of the discriminant in  $x^2 + 2x + 1 = 0$ ?

- (A) 0  
 (B) 1  
 (C) 2  
 (D) 4  
 (E) 8

## Open Ended Questions (FRQ)

**Q9.** Solve  $2x^2 - 5x - 3 = 0$  using the quadratic formula. Show all work and explain your reasoning.

**Q10.** Find the value of the discriminant in  $x^2 - 9x + 20 = 0$ . Explain what this means for the number and type of solutions.

## Chef's Tips

- Check calculations carefully
- Make sure to show work for FRQs
- Consider using a calculator for complex calculations